# Operating Temperature Range

The allowable operating temperature range for needle roller bearings is generally -20～+120°C.
When operating at temperatures outside this range, the operation may be limited by the allowable temperature range of prepacked grease, seal, cage material, etc.
The operating temperature range for some types of bearings is different from the above. See the section for each bearing.

# Handling of Bearings

## Precautions in handling

Since the bearing is a high-accuracy mechanical element, special attention must be paid to its handling. The following precautions should be noted when handling the bearings.

**❶ Bearings and their surrounding parts should be kept clean.**
Bearings and their surrounding parts must be kept clean paying special attention to dust and dirt. Tools and the working environment should also be cleaned.

**❷ Bearings should be handled carefully.**
A shock load during handling may cause scratches, indentations and even cracks or chips on the raceway surfaces and rolling elements.

**❸ Bearings should be mounted or dismounted with proper tools.**
When mounting and dismounting, tools suitable for the bearing type should be used.

**❹ Bearings should be protected against corrosion.**
Bearings are treated with anti-corrosive oil. However, when handling them with bare hands, sweat from the hands may result in future rust formation. Gloves should be worn, or hands should be dipped in mineral oil.

## Mounting

**Preparation**

Before mounting the bearing, the dimensions and fillets of the shaft and housing should be checked to ensure that they conform to specifications.
Bearings should be unwrapped just before mounting. In case of grease lubrication, bearings should be filled with grease without cleaning the bearings. Even in the case of oil lubrication, it is normally unnecessary to clean the bearings. However, when high accuracy is required or when using at high speeds, the bearings should be cleaned using cleaning oil to remove thoroughly oily contents. The cleaned bearings should not be left alone without anti-corrosive precautions, because bearings can easily be corroded after anti-corrosive agents are removed.
Lubricating grease is prepacked in some types of bearings. Therefore, refer to the relevant section for each bearing.





**Methods of mounting**

Mounting methods of bearings are different according to the type of bearing and the fit. In general, mounting of needle roller bearings is comparatively easy. However, non-separable bearings with large interferences should be handled with great care.

**❶ Mounting by press fit**
Small and medium bearings with small interferences require a small pressing-in force for mounting, and they are mounted using a press at room temperature. The bearing should be pressed in carefully, applying a force evenly to the bearing with a fitting tool as shown in Fig. 26. For separable bearings, the inner and outer rings can be mounted separately, and the mounting work is simple. However, when installing the shaft and inner ring assembly into the outer ring, care should be taken not to damage the raceway surfaces and rolling elements.

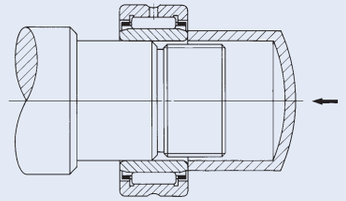


**Fig. 26**

When mounting non-separable bearings, the inner and outer rings are pressed in simultaneously by applying a cover plate as shown in Fig. 27. It must never happen that the inner ring is press-fitted to the shaft by striking the outer ring, or the outer ring by striking the inner ring, because the raceway surfaces and rolling elements will be scratched or indented.

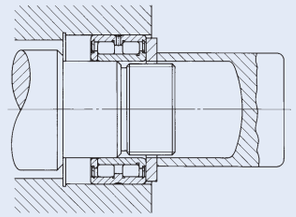


**Fig. 27**

When press fitting, the friction of the fitting surfaces can be reduced by applying high viscosity oil over the fitting surfaces.
The pressing-in or pulling-out force to be applied to the bearing is given on page A59.

**❷ Mounting by shrink fitting**
This method is used when the interference is great or when a large bearing is to be fitted. The housing is heated and thermally expanded when fitting the outer ring to the housing and the inner ring is heated and expanded when fitting it to the shaft allowing the bearing to be set easily within a short time. The maximum allowable temperature for the shrink fit is +120°C, and heating should be performed appropriately. Pure non-corrosive mineral oil is recommended as the heating oil for shrink fit, and insulation oil for transformers is considered to be the best. During cooling, the bearing also shrinks in the axial direction. Therefore, to ensure that there is no clearance between the bearing and the shoulder, an axial force must be applied continuously to the bearing until it has cooled.
When the interference between the outer ring and the housing is great, an expansion fit method in which the bearing is cooled using dry ice or other cooling agent before fitting can be used. Immediately after fitting, however, moisture from the air easily condenses on the bearing. Therefore, it is necessary to take preventive measures against corrosion.



1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

## Pressing force and pulling force

Guidelines for the pressing force when pressing in the inner ring to the shaft and the pulling force when pulling it out are obtained from the following equation.

$$K = f_k \frac{d}{d+2} \Delta_{df} B \left\{1 - \left(\frac{d}{F}\right)^2\right\} \quad \cdots\cdots\cdots(39)$$

where, $K$ : Pressing or pulling force, N
$f_k$ : Resistance factor determined by the coefficient of friction
When pressing in inner ring to shaft, $f_k = 4 \times 10^4$
When pulling out inner ring from shaft, $f_k = 6 \times 10^4$
$d$ : Bore diameter of inner ring, mm
$\Delta_{df}$ : Apparent interference, mm
$B$ : Width of inner ring, mm
$F$ : Outside diameter of inner ring, mm

The actual pressing force or pulling force may be greater than the calculated value due to mounting errors. When designing a puller, it is necessary that the puller has the strength (rigidity) to withstand more than 5 times the calculated value.

## Running test

After mounting the bearing, a running test is carried out to check whether the mounting is normal. Usually, it is first checked by manual turning. Then, it is operated by power gradually from no-load and low-speed up to normal operating conditions to check for abnormalities.
Noise can be checked by using a soundscope or similar instrument. In this test, checks are carried out for the following abnormalities.

**❶ Manual turning**
(a) Uneven torque ····· Improper mounting
(b) Sticking and rattling ··· Scratches or indentations on the raceway surface
(c) Irregular noise ··· Penetration of dust or foreign particles

**❷ Power running**
(a) Abnormal noise or vibration ··· Indentations on the raceway surface, too great clearance
(b) Abnormal temperature ··· Unsuitable lubricant, improper mounting, too small clearance

## Dismounting

Dismounting of the bearings is carried out for the periodic inspection or repairs of machines. By inspecting the bearing, related parts or mechanisms, lubrication, etc., important data is obtained. In the same manner as in mounting, care should be taken to prevent damage to the bearing or other parts.
A suitable dismounting method should be selected according to the type of the bearing, fit, etc. Bearings mounted by interference fit are especially difficult to dismount, and it is necessary to give due consideration to the structure around the bearing during the design stage.

### Dismounting of outer ring

Outer rings mounted by interference fit are dismounted as shown in Fig. 28, by screwing in the push-out bolts evenly through several screw holes provided at places corresponding to the side face of the outer ring.

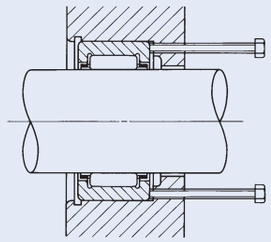


Fig. 28

### Dismounting of inner ring

In the case of bearings such as needle roller bearings in which the inner and outer rings are separable, the simplest way to press out the inner ring is by using a press as shown in Fig. 29.
The puller shown in Fig. 30 is also generally used. This is designed according to the bearing size. In addition, there are a 3-hook puller (Fig. 31) and a 2-hook puller for wide-range use.

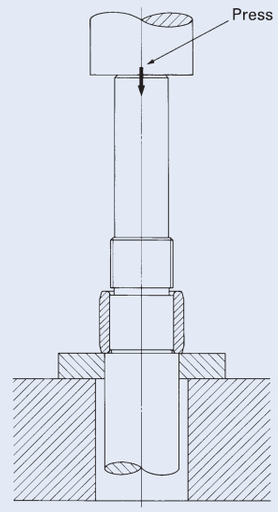



Fig. 29

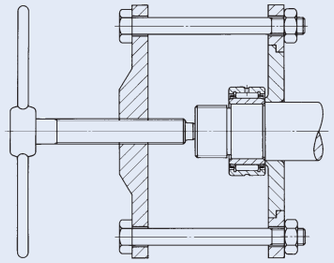


Fig. 30

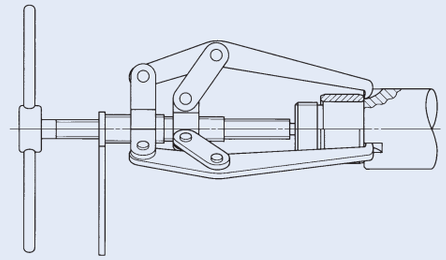


Fig. 31

In addition to these, when it is difficult to remove the inner ring due to high shoulders, several holes for removal pins are made through the shoulder, or several hook grooves are cut in the shoulder as shown in Fig. 32 and Fig. 33.
When a bearing is not to be used again after removal, it may be removed by heating with a torch lamp.

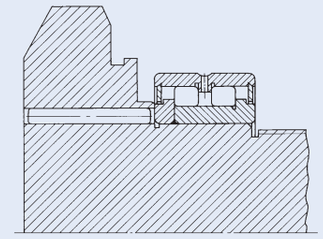


Fig. 32

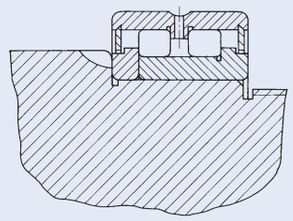


Fig. 33





1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

## Inspection of bearing

### Cleaning of bearing

When inspecting a bearing after removal, the appearance of the bearing should be recorded first. Then, after the residual amount of lubricant is checked and a sample of lubricant is collected, the bearing should be cleaned.
For cleaning, light oil or kerosene is commonly used. Cleaning is divided into rough cleaning and final cleaning, and wire gauze is set as a raised bottom in a container to prevent the bearing from touching the bottom of the container.
Lubricating grease and adhering substances such as foreign particles are removed with a brush, etc., using oil for rough cleaning. Care should be taken during this process, because if the bearing is turned with foreign particles attached, the raceway surfaces may be scratched.
Final cleaning is carried out by turning the bearing in cleaning oil. It is desirable that the cleaning oil is kept clean by filtering. Immediately after cleaning, the bearing must be protected against corrosion.

### Inspection and evaluation of bearing

The judgement as to whether the removed bearing is reusable depends on the inspection after cleaning. Conditions of the raceway surfaces, rolling elements and fitting surfaces, wear condition of the cage, increase of bearing clearance, dimensions, rotational accuracy, etc. should be checked for damage and abnormalities.
The evaluation is performed based on the experience taking into consideration the degree of damage, machine performance, importance of the machine, operating conditions, period until the next inspection, and other such factors.

## Maintenance and inspection

### Maintenance and inspection

Maintenance and inspection are carried out to maintain good performance of bearings installed in the machine.
Maintenance is performed by checking the machine operating conditions, checking and replenishing or replacing the lubricant, checking the bearing and related parts by periodic disassembly and other such procedures.
Items for inspection of a running bearing in a machine include the bearing temperature, noise, vibration and condition of lubricant.
When any abnormality is found during operation, the cause should be investigated and measures taken by referring to the section on running test on page 62.
When removing a bearing, refer to the section on dismounting on page A59.

### Damage, causes and corrective action

Rolling bearings can generally be used fully up to their rolling fatigue life if they are properly selected, mounted, operated and maintained. However, they may actually be damaged earlier than their expected lifetimes creating problems or accidents. Common causes of damage include improper mounting or handling, insufficient lubrication and penetration of foreign particles.
It may be difficult to determine the exact cause of a problem by checking only the damaged bearing. The conditions of the machine before and after the occurrence of the damage, the location and the operating and ambient conditions of the bearing, the structure around the bearing, etc. should also be examined. It then becomes possible to assess the cause of the damage by linking the conditions of the damaged bearing to the probable causes arising from the machine operation, and to prevent the recurrence of similar problems.
Common types of damage, causes and corrective action are listed in Table 41.





**Table 41 Damage, causes and corrective action**

| Condition of bearing damage | | Cause | Corrective action |
|---|---|---|---|
| Flaking | Flakings at opposite circumferential positions on raceway surfaces | Improper roundness of housing bore | Correction of housing bore accuracy |
| | Flakings in the vicinity of raceway surface edges and roller ends | Improper mounting, Shaft deflection, Poor centering, Poor accuracy of shaft or housing | Careful mounting, Careful centering, Correction of shoulders of shaft and housing for right angles |
| | Flakings on raceway surfaces with an interval corresponding to roller pitch | Great shock load when mounting, Rusting during machine stoppage | Careful mounting, Protection against rust for long periods of machine stoppage |
| | Early flaking on raceway surfaces and rolling elements | Too small clearance, Too great load, Poor lubrication, Rusting, etc. | Correct selection of fit and clearance<br>Correct selection of lubricant |
| Galling | Galling on raceway surfaces and rolling surfaces of rollers | Poor lubrication in early stage<br>Grease consistency too hard<br>High acceleration at start | Selection of softer grease,<br>Avoiding quick acceleration |
| | Galling between roller end faces and collar guide surfaces | Poor lubrication, Poor mounting,<br>Large axial load | Correct selection of lubricant<br>Correct mounting |
| Breakage | Cracks in outer or inner ring | Excessive shock load, Too much interference.<br>Poor cylindricity of shaft. Too large fillet radius,<br>Development of thermal cracks,<br>Development of flaking | Reevaluation of load conditions, Correction of fit,<br>Correction of machining accuracy of shaft or sleeve, Making fillet radius smaller than the chamfer dimension of bearing |
| | Cracked rolling elements,<br>broken collar | Development of flaking<br>Shock to collar when mounting,<br>Dropped by careless handling | Careful handling and mounting |
| | Broken cage | Abnormal load to cage by poor mounting,<br>Poor lubrication | Minimizing mounting errors,<br>Study of lubricating method and lubricant |
| Dent | Indentations on raceway surfaces at an interval corresponding to the pitch between rolling elements (brinelling) | Shock load applied when mounting,<br>Excessive load while stopping | Careful handling |
| | Indentation on raceway surfaces and rolling surfaces of rollers | Biting of foreign substances such as metal chips and sands | Cleaning of housing, Improvement of sealing,<br>Use of clean lubricant |
| Abnormal wear | False brinelling<br>(Phenomenon like brinelling) | Vibration when the bearing is stationary such as during transportation,<br>Oscillating motion with small amplitude | Fixing of shaft and housing,<br>Use of lubricating oil,<br>Application of preload to reduce vibration |
| | Fretting<br>Localized wear of fitted surfaces accompanied by red-brown wear particles | Sliding between fitted surfaces | Increase of interference,<br>Application of oil |
| | Wear on raceway surfaces, collar surfaces, rolling surfaces of rollers, cages, etc. | Penetration of foreign particles, Poor lubrication, Rust | Improvement of sealing, Cleaning of housing<br>Use of clean lubricant |
| | Creep<br>Wear on fitted surfaces | Sliding between fitted surfaces,<br>Insufficient tightening of sleeve | Increase of interference,<br>Correct tightening of sleeve |
| Seizure | Discoloration of rolling elements and/or raceway surfaces and/or flange surfaces, Adhesion and welding, Discoloration of cage | Poor lubrication, Too small clearance, Poor mounting | Supply of proper amount of proper lubricant,<br>Rechecking of fit and bearing clearance<br>Rechecking of mounting dimensions and related parts |
| Electric corrosion | Ripples on raceway surfaces | Melting by sparks due to electric current | Insulation of bearing,<br>Grounding to avoid electric current |
| Rust, corrosion | Rust or corrosion on bearing inside surfaces or on fitted surfaces | Condensation of vapor in air,<br>Penetration of corrosive substances | Careful storage if under high temperature and high humidity, Protection against rust,<br>Improvement of sealing |



1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

IKO

# Description of Each Series & Table of Dimensions


**Shell Type Needle Roller Bearings** TA・TLA・BA・BHA ..... B1

**Needle Roller Cages** for general usage KT ..... C1

**Needle Roller Cages** for engine connecting rods KT…EG・KTV…EG ..... C17

**Machined Type Needle Roller Bearings** TAF…/SG・NA・TAFI・TRI・BRI ..... D1

**Needle Roller Bearings** with separable cage NAF ..... D93

**Roller Bearings** NAG・NAU・TRU・NAS ..... E1

**Thrust Bearings** NTB・AS・AZK・WS・GS ..... F1

**Combined Type Needle Roller Bearings** NAX・NBX・NATA・NATB ..... G1

**Inner Rings** IRT・IRB・LRT・LRB ..... H1

**Cam Followers** CF…/SG・CF・NUCF・CFS・CR ..... I1

**Roller Followers** NAST・NART・NURT ..... I83

**Crossed Roller Bearings** CRBF・CRBH・CRBC・CRB・CRBS ..... J1

**Spherical Bushings** SB・GE・SBB ..... K1

**Pilloballs** PB・PHS・POS・PHSB・POSB・PHSA ..... K29

**L-balls** LHSA・LHS ..... K45

**Super Flexible Nozzles** SNA・SNM・SNPT ..... K55

**Parts For Needle Roller Bearings** OS・DS・WR・AR・Needle Roller ..... L1




# SHELL TYPE NEEDLE ROLLER BEARINGS

- **Shell Type Caged Needle Roller Bearings**
- **Shell Type Grease Retained Full Complement Needle Roller Bearings**










## Structure and features

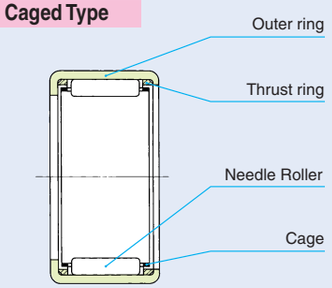
IKO Shell Type Needle Roller Bearings are lightweight bearings with large load ratings. They employ a shell type outer ring made from a thin special-steel plate which is accurately drawn, carburized and quenched, thus providing the lowest sectional height among the needle roller bearings.

There are two types of bearings available in this series; the caged type and the full complement type. The appropriate type can be selected according to the operating conditions. The caged type has a structure in which the needle rollers are accurately guided by the cage and thrust rings. It is useful for applications at high-speed rotation. The full complement type needle roller bearing, on the other hand, is suitable for heavy-load applications at low-speed rotation.

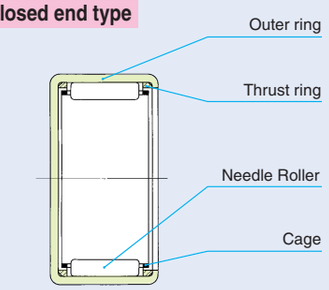
Since these bearings are press-fitted into the housing, no fixtures for axial positioning are needed. They are ideal for use in mass-produced articles that require economy, and have a wide variety of applications.

### Structures of Shell Type Needle Roller Bearings

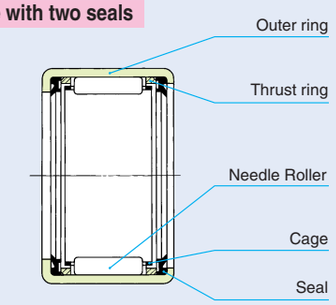
**Standard Caged Type**



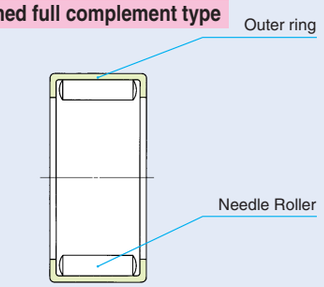

**Caged and closed end type**




**Caged type with two seals**




**Grease retained full complement type**



# Types

Numerous varieties of Shell Type Needle Roller Bearings are available as shown in Table 1.

**Table 1 Type of bearing**

| Series | | Type: Caged – Standard | Caged – Closed end | Caged – With seals (1) | Full complement – Grease retained |
|---|---|---|---|---|---|
| Metric series | — | TLA ···Z | TLAM | TLA···UU | YTL |
| | Heavy duty | TA ···Z | TAM | — | YT |
| Inch series | — | BA ···Z | BAM | — | YB |
| | Heavy duty | BHA ···Z | BHAM | — | YBH |

Note(1) When the heavy duty type with seals or the closed end type with one seal is required, please consult IKO.

Remark A "W" is added to the model code to indicate that the rolling elements are of the double-row type.
Example TAW 5045 Z

## Shell Type Caged Needle Roller Bearings

### Standard type

This type has a narrow gap between the bore of the marked-side flange of the outer ring (brand, bearing number, etc. are marked) and the shaft, which prevents grease leaks and the entry of foreign particles. This type has wide applications.

### Closed end type

This type is completely closed on one side of the outer ring, and is ideal for use when perfect closing of shaft ends is desired.

The shape of the closed end surface of the outer ring is divided into two types, and the dimensions $t_1$ and $t_2$ in the illustrations shown in the dimension tables apply to the bearings with the roller set bore diameters, $F_w > 22$ and $F_w \leqq 22$, respectively.

### Type with seals at both sides

This type has a wider outer ring than the standard type and is installed with seals consisting of a reinforcing ring and special synthetic rubber to prevent grease leaks and the entry of foreign particles.

## Shell Type Grease Retained Full Complement Needle Roller Bearings

This type has full complement rollers which extend to the full width of the outer ring raceway. It can, therefore, withstand heavy bearing loads and is most suitable for low and medium rotational speeds as well as rocking motions. As lubricating grease is prepacked with the rollers, the bearing can be operated immediately after being fitted.

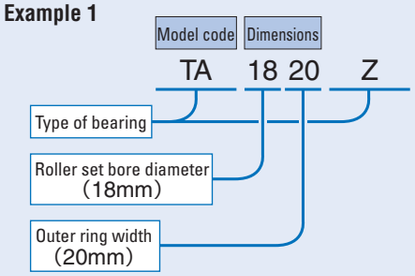
# Identification Number

The identification number of Shell Type Needle Roller Bearings consists of a model code and dimensions. Examples of the arrangement are shown below.

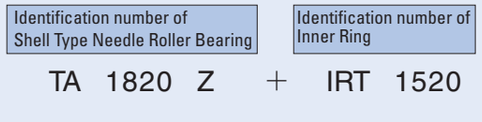
When using with inner rings, the assembled inner rings shown in the dimension tables are used. An example in this case is also shown below. Inner rings are delivered separately.

### Examples of identification number

**Example 1**




**Example 2 (With inner ring)**




# Accuracy

The outer rings of Shell Type Needle Roller Bearings are thin and therefore cannot avoid deformation due to heat treatment. It is thus not appropriate to take direct measurements of the bearing. The roller set bore diameter is measured using a plug gauge or tapered gauge after press-fitting the bearing to a suitable ring gauge. The gauge specifications are shown in Tables 2.1 and 2.2.

Tolerances of outer ring width $C$ are shown in Table 3.







**Table 2.1 Measuring gauges for metric series bearings** unit: mm

| $F_w$ Nominal roller set bore diameter | Ring gauge TA···Z(1) | Ring gauge TLA···Z(2) | Plug gauge Go | Plug gauge No-go |
|---|---|---|---|---|
| 4 | — | 7.981 | 4.004 | 4.016 |
| 5 | — | 8.981 | 5.004 | 5.016 |
| 6 | — | 9.981 | 6.004 | 6.016 |
| 7 | — | 10.977 | 7.005 | 7.020 |
| 8 | 14.992 | 11.977 | 8.005 | 8.020 |
| 9 | 15.992 | 12.977 | 9.005 | 9.020 |
| 10 | 16.992 | 13.977 | 10.005 | 10.020 |
| 12 | 18.991 | 15.977(3)<br>17.977(3) | 12.006 | 12.024 |
| 13 | — | 18.972 | 13.006 | 13.024 |
| 14 | 21.991 | 19.972 | 14.006 | 14.024 |
| 15 | 21.991 | 20.972 | 15.006 | 15.024 |
| 16 | 23.991 | 21.972 | 16.006 | 16.024 |
| 17 | 23.991 | 22.972 | 17.006 | 17.024 |
| 18 | 24.991 | 23.972 | 18.006 | 18.024 |
| 19 | 26.991 | — | 19.007 | 19.028 |
| 20 | 26.991(4)<br>27.991(4) | 25.972 | 20.007 | 20.028 |
| 21 | 28.991 | — | 21.007 | 21.028 |
| 22 | 28.991(5)<br>29.991(5) | 27.972 | 22.007 | 22.028 |
| 24 | 30.989(6)<br>31.989(6) | — | 24.007 | 24.028 |
| 25 | 32.989 | 31.967 | 25.007 | 25.028 |
| 26 | 33.989 | — | 26.007 | 26.028 |
| 28 | 36.989 | 34.967 | 28.007 | 28.028 |
| 29 | 37.989 | — | 29.007 | 29.028 |
| 30 | 39.989 | 36.967 | 30.007 | 30.028 |
| 32 | 41.989 | — | 32.009 | 32.034 |
| 35 | 44.989 | 41.967 | 35.009 | 35.034 |
| 37 | 46.989 | — | 37.009 | 37.034 |
| 38 | 47.989 | — | 38.009 | 38.034 |
| 40 | 49.989 | 46.967 | 40.009 | 40.034 |
| 45 | 54.988 | 51.961 | 45.009 | 45.034 |
| 50 | 61.988 | 57.961 | 50.009 | 50.034 |
| 55 | 66.988 | 62.961 | 55.010 | 55.040 |
| 60 | 71.988 | — | 60.010 | 60.040 |
| 62 | 73.988 | — | 62.010 | 62.040 |
| 65 | 76.988 | — | 65.010 | 65.040 |
| 70 | 81.987 | — | 70.010 | 70.040 |

Notes(1) Also applicable to TAM and YT
(2) Also applicable to TLAM, YTL, TLA···UU
(3) The upper value is for TLA 1210Z model, and the lower value is for TLA 1212Z model.
(4) The lower value is for TA 202820Z model, and the upper value is for models other than TA 202820Z model.
(5) The lower value is for TA 223016Z and TA 223020Z models, and the upper value is for models other than those models.
(6) The lower value is for TA 243216Z and TA 243220Z models, and the upper value is for models other than those models.

**Table 2.2 Measuring gauges for inch series bearings** unit: mm

| $F_w$ Nominal roller set bore diameter | Ring gauge BA···Z(1) | Ring gauge BHA···Z(2) | Plug gauge Go | Plug gauge No-go |
|---|---|---|---|---|
| 3.969 | 7.155 | — | 3.990 | 4.016 |
| 4.762 | 8.730 | — | 4.783 | 4.808 |
| 6.350 | 11.125 | — | 6.388 | 6.414 |
| 7.938 | 12.713 | 14.300 | 7.976 | 8.001 |
| 9.525 | 14.300 | 15.888 | 9.563 | 9.588 |
| 11.112 | 15.888 | 17.475 | 11.151 | 11.176 |
| 12.700 | 17.475 | 19.063 | 12.738 | 12.764 |
| 14.288 | 19.063 | 20.650 | 14.326 | 14.351 |
| 15.875 | 20.650 | 22.238 | 15.913 | 15.938 |
| 17.462 | 22.238 | 23.825 | 17.501 | 17.526 |
| 19.050 | 25.387 | 26.975 | 19.063 | 19.088 |
| 20.638 | 26.975 | 28.562 | 20.650 | 20.676 |
| 22.225 | 28.562 | 30.150 | 22.238 | 22.263 |
| 23.812 | 30.150 | — | 23.825 | 23.851 |
| 25.400 | 31.737 | 33.325 | 25.413 | 25.438 |
| 26.988 | 33.325 | — | 27.000 | 27.026 |
| 28.575 | 34.912 | 38.087 | 28.588 | 28.613 |
| 30.162 | 38.087 | — | 30.175 | 30.201 |
| 31.750 | 38.087 | 41.262 | 31.763 | 31.788 |
| 33.338 | 41.262 | — | 33.350 | 33.378 |
| 34.925 | 41.262 | 44.437 | 34.938 | 34.966 |
| 38.100 | 47.612 | — | 38.113 | 38.143 |
| 41.275 | 50.787 | — | 41.288 | 41.318 |
| 44.450 | 53.962 | 57.137 | 44.463 | 44.496 |
| 47.625 | 57.137 | — | 47.638 | 47.671 |
| 50.800 | 60.312 | — | 50.815 | 50.848 |
| 52.388 | — | 64.280 | 52.413 | 52.451 |
| 53.975 | 63.487 | — | 53.990 | 54.028 |
| 57.150 | 66.662 | — | 57.165 | 57.203 |
| 66.675 | 76.187 | — | 66.700 | 66.738 |
| 69.850 | 79.362 | — | 69.875 | 69.914 |

Notes(1) Also applicable to BAM and YB
(2) Also applicable to BHAM and YBH

**Table 3 Tolerances of outer ring width $C$** unit: mm

| Series | Tolerance |
|---|---|
| Metric | 0 ～ −0.20 |
| Inch | 0 ～ −0.25 |

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

## Fit

As the outer ring is thin, the correct dimensions and accuracy of Shell Type Needle Roller Bearings are obtained only after they have been press-fitted into the housing bore. Bearing accuracy is directly affected by housing dimensions, shape and rigidity. This should be taken into account when considering fit and accuracy. The radial clearance after fitting the bearing to the shaft and the housing bore varies with their tolerances.

Table 4 shows the recommended fit for Shell Type Needle Roller Bearings.

Table 5 shows a calculation example of radial clearance after fitting. This calculation applies to bearings without inner ring to be fitted into rigid steel or cast iron housings. When the housing is made of light alloy or a thin steel pipe, it is necessary to check dimensions by actual measurement.

Generally, when making the radial clearance smaller, it is recommended that the shaft diameter be increased, without decreasing the housing bore diameter.

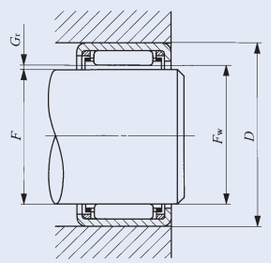
**Table 4 Recommended fit**

| Type of bearing | Housing material | Tolerance class | | |
|---|---|---|---|---|
| | | Shaft (1) | | Housing bore |
| | | Without inner ring | With inner ring | |
| TA···Z, BA···Z, BHA···Z, TAM, BAM, BHAM, YT, YB, YBH | Steel Cast iron | h6 | k5(j5) | J7 |
| | Light alloy (Thin steel pipe) | h6 | k5(j5) | M7(N7) |
| TLA···Z, TLAM, YTL, TLA···UU | Steel Cast iron | h6 | k5(j5) | N7 |
| | Light alloy (Thin steel pipe) | h6 | k5(j5) | R7(S7) |

Note(1) When housings are made of light alloy or a thin steel pipe, the roller set bore diameter is greatly affected by the housing thickness and shape. Therefore, before mass-production assembly, assembly tests should be carried out to confirm the amount of dimensional change and to determine the tolerance of the shaft which will give normal clearances.

**Table 5 Calculation example of radial clearance after fitting**

unit: mm




$D$ : Housing bore diameter
$F_w$ : Roller set bore diameter
$F$ : Shaft diameter
$G_r$ : Radial clearance

| Calculation procedure | Example of TLA 2020 Z |
|---|---|
| ❶ Dimension of roller set bore diameter of bearing after it has been press-fitted into the ring gauge.<br>Dimension of ring gauge ($D_0$): See Tables 2.1 and 2.2 on page B4.<br>Max. value of roller set bore dia. ($F_{w\,max}$): No-go dimension of plug gauge<br>Min. value of roller set bore dia. ($F_{w\,min}$): Go dimension of plug gauge | From Table 2.1 on page B4<br>$D_0 = 25.972$<br>$F_{w\,max} = 20.028$<br>$F_{w\,min} = 20.007$ |
| ❷ Dimension of housing bore<br>Max. value of housing bore ($D_{max}$): See the dimension table.<br>Min. value of housing bore ($D_{min}$): See the dimension table. | From the dimension table on page B14,<br>$D_{max} = 25.993$<br>$D_{min} = 25.972$ |
| ❸ Dimension of roller set bore diameter of bearing after it has been press-fitted into the housing bore<br>Max. value of roller set bore dia.($F_{we\,max}$)=($D_{max} - D_0$) + $F_{w\,max}$<br>Min. value of roller set bore dia.($F_{we\,min}$)=($D_{min} - D_0$) + $F_{w\,min}$ | From the equations,<br>$F_{we\,max} = 20.049$<br>$F_{we\,min} = 20.007$ |
| ❹ Dimension of shaft<br>Max. value of shaft dia. ($F_{max}$): See the dimension table.<br>Min. value of shaft dia. ($F_{min}$): See the dimension table. | From the dimension table on page B14,<br>$F_{max} = 20.000$<br>$F_{min} = 19.987$ |
| ❺ Radial clearance after mounting<br>Max. value of radial clearance($G_{r\,max}$)= $F_{we\,max} - F_{min}$<br>Min. value of radial clearance($G_{r\,min}$)= $F_{we\,min} - F_{max}$ | From the equations,<br>$G_{r\,max} = 0.062$<br>$G_{r\,min} = 0.007$<br>The radial clearance after mounting becomes 0.007~0.062 mm. |







## Lubrication

Bearings with prepacked grease are shown in Table 6. ALVANIA GREASE S2 (SHELL) is prepacked as the lubricating grease.

In the case of bearings without prepacked grease, perform proper lubrication for use. If the bearings are operated without lubrication, the wear of the roller contact surfaces will increase and the bearing life will be shortened.

## Oil Hole

For Shell Type Needle Roller Bearings with an oil hole, "OH" is appended to the end of the identification number.

**Example** TA 2525 Z OH

The symbol "OH" is not marked on the bearing itself, but is shown on its packaging, etc. When bearings with multiple oil holes are required, please consult IKO.

**Table 6 Bearings with prepacked grease**

○ : With prepacked grease × : Without prepacked grease

| Series \ Bearing type | | Caged | | | Full complement |
|---|---|---|---|---|---|
| | | Standard | Closed end | With seals | Grease retained |
| Metric series | TLA, TLAM, YTL | × | × | ○ | ○ |
| | TA, TAM, YT | × | × | – | ○ |
| Inch series | BA, BAM, YB | × | × | – | ○ |
| | BHA, BHAM, YBH | × | × | – | ○ |

## Static Safety Factor

Since Shell Type Needle Roller Bearings employ an outer ring made from a thin steel plate which is drawn, carburized and quenched, excessively large loads must be avoided. The required static safety factor is usually more than 3.

## Specifications of shaft and housing

Shell Type Needle Roller Bearings are commonly used without an inner ring. In such cases, the surface hardness of the raceway surface should be 58～64HRC and the surface roughness should not exceed 0.2 $\mu$m$R_a$. However, when the operating condition is not severe, a surface roughness 0.8 $\mu$m$R_a$ or less can be used.

If the surface hardness is low, the load rating must be corrected by the hardness factor shown on page A20. When the shaft cannot be heat treated and finished by grinding, the use of IKO Inner Rings for Shell Type Needle Roller Bearings (See page H1.) is recommended.

## Mounting

Shell Type Needle Roller Bearings should be pressed into the housings gently using the appropriate tool as shown in Fig. 1, with their marked end surface up. As the outer ring is thin, it must never be struck directly with a hammer.

Since the outer rings of Shell Type Needle Roller Bearings are firmly fitted to housing bores with interference, it is unnecessary to fix them axially. Fig. 2 shows mounting examples.

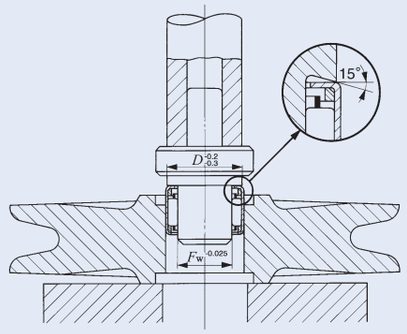



**Fig.1 Example of mounting tool**

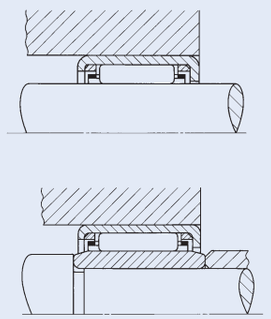


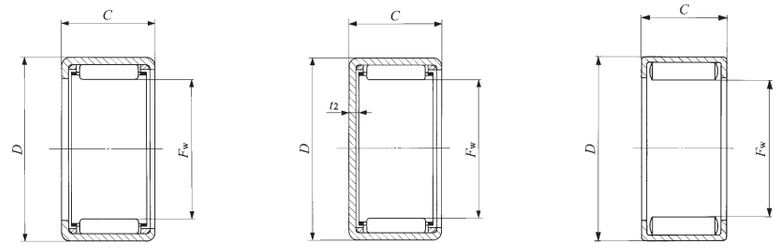
**Fig.2 Mounting examples**

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

# SHELL TYPE NEEDLE ROLLER BEARINGS








TA···Z TLA···Z　　TAM TLAM　　YT YTL

## Shaft dia. 4 — 10mm

| Shaft dia. mm | Identification number | | | | | | | | | | Boundary dimensions mm | | | | Standard mounting dimensions mm | | | | | | Basic dynamic load rating $C$ N | Basic static load rating $C_0$ N | Allowable rotational speed(1) rpm | Assembled inner ring |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Standard | Mass (Ref.) g | Closed end | Mass (Ref.) g | Standard | Mass (Ref.) g | Closed end | Mass (Ref.) g | Grease retained | Mass (Ref.) g | $F_w$ | $D$ | $C$ | $t_2$ Max. | Shaft dia. h6 Max. | Shaft dia. h6 Min. | Housing bore dia. J7 Max. | Housing bore dia. J7 Min. | Housing bore dia. N7 Max. | Housing bore dia. N7 Min. | | | | |
| 4 | — | — | — | — | TLA 48 Z | 1.54 | TLAM 48 | 1.67 | — | — | 4 | 8 | 8 | 1 | 4.000 | 3.992 | — | — | 7.996 | 7.981 | 1 350 | 1 010 | 75 000 | — |
| | — | — | — | — | — | — | — | — | YTL 48 | 1.73 | 4 | 8 | 8 | — | | | | | | | 3 010 | 2 900 | 40 000 | — |
| 5 | — | — | — | — | TLA 59 Z | 1.9 | TLAM 59 | 2 | — | — | 5 | 9 | 9 | 1 | 5.000 | 4.992 | — | — | 8.996 | 8.981 | 1 880 | 1 600 | 65 000 | — |
| | — | — | — | — | — | — | — | — | YTL 59 | 2.4 | 5 | 9 | 9 | — | | | | | | | 4 320 | 4 750 | 30 000 | — |
| 6 | — | — | — | — | TLA 69 Z | 2.2 | TLAM 69 | 2.3 | — | — | 6 | 10 | 9 | 1 | 6.000 | 5.992 | — | — | 9.996 | 9.981 | 2 100 | 1 900 | 55 000 | — |
| 7 | — | — | — | — | TLA 79 Z | 2.5 | TLAM 79 | 2.7 | — | — | 7 | 11 | 9 | 1 | 7.000 | 6.991 | — | — | 10.995 | 10.977 | 2 490 | 2 450 | 50 000 | — |
| 8 | — | — | — | — | TLA 810 Z | 3.1 | TLAM 810 | 3.3 | — | — | 8 | 12 | 10 | 1 | 8.000 | 7.991 | — | — | 11.995 | 11.977 | 3 320 | 3 670 | 45 000 | — |
| | TA 810 Z | 6.7 | TAM 810 | 7.1 | — | — | — | — | — | — | 8 | 15 | 10 | 1.3 | 8.000 | 7.991 | 15.010 | 14.992 | — | — | 3 470 | 2 880 | 45 000 | — |
| | TA 815 Z | 9.7 | TAM 815 | 10.1 | — | — | — | — | — | — | 8 | 15 | 15 | 1.3 | | | | | | | 5 780 | 5 570 | 45 000 | — |
| | TA 820 Z | 12.9 | TAM 820 | 13.3 | — | — | — | — | — | — | 8 | 15 | 20 | 1.3 | | | | | | | 8 340 | 8 920 | 45 000 | — |
| | — | — | — | — | — | — | — | — | YT 810 | 7.7 | 8 | 15 | 10 | — | | | | | | | 7 530 | 7 950 | 19 000 | — |
| 9 | — | — | — | — | TLA 910 Z | 3.4 | TLAM 910 | 3.6 | — | — | 9 | 13 | 10 | 1 | 9.000 | 8.991 | — | — | 12.995 | 12.977 | 3 500 | 4 040 | 45 000 | — |
| | — | — | — | — | TLA 912 Z | 4 | TLAM 912 | 4.3 | — | — | 9 | 13 | 12 | 1 | | | | | | | 4 460 | 5 510 | 45 000 | — |
| | TA 912 Z | 8.7 | TAM 912 | 9.2 | — | — | — | — | — | — | 9 | 16 | 12 | 1.3 | 9.000 | 8.991 | 16.010 | 15.992 | — | — | 5 140 | 4 880 | 45 000 | — |
| | TA 916 Z | 11.4 | TAM 916 | 11.9 | — | — | — | — | — | — | 9 | 16 | 16 | 1.3 | | | | | | | 6 960 | 7 210 | 45 000 | — |
| | — | — | — | — | — | — | — | — | YT 912 | 10.1 | 9 | 16 | 12 | — | | | | | | | 9 690 | 11 200 | 17 000 | — |
| 10 | — | — | — | — | TLA 1010 Z | 3.7 | TLAM 1010 | 4 | — | — | 10 | 14 | 10 | 1 | 10.000 | 9.991 | — | — | 13.995 | 13.977 | 3 870 | 4 740 | 40 000 | IRT 710 |
| | — | — | — | — | TLA 1012 Z | 4.4 | TLAM 1012 | 4.8 | — | — | 10 | 14 | 12 | 1 | | | | | | | 4 920 | 6 460 | 40 000 | IRT 712 |
| | — | — | — | — | TLA 1015 Z | 5.5 | TLAM 1015 | 5.9 | — | — | 10 | 14 | 15 | 1 | | | | | | | 6 390 | 9 040 | 40 000 | IRT 715 |
| | TA 1010 Z | 7.9 | TAM 1010 | 8.5 | — | — | — | — | — | — | 10 | 17 | 10 | 1.3 | 10.000 | 9.991 | 17.010 | 16.992 | — | — | 4 150 | 3 780 | 40 000 | IRT 710 |
| | TA 1012 Z | 9.3 | TAM 1012 | 10 | — | — | — | — | — | — | 10 | 17 | 12 | 1.3 | | | | | | | 5 590 | 5 540 | 40 000 | IRT 712 |
| | TA 1015 Z | 11.5 | TAM 1015 | 12.2 | — | — | — | — | — | — | 10 | 17 | 15 | 1.3 | | | | | | | 6 920 | 7 300 | 40 000 | IRT 715 |
| | TA 1020 Z | 15.4 | TAM 1020 | 16 | — | — | — | — | — | — | 10 | 17 | 20 | 1.3 | | | | | | | 9 990 | 11 700 | 40 000 | — |

Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remark Shell Type Grease Retained Full Complement Needle Roller Bearings are provided with prepacked grease. Standard type and closed end type bearings are not provided with prepacked grease, so perform proper lubrication when using these types of bearings.

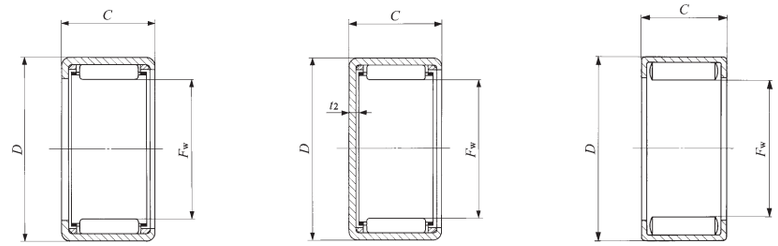


1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

# SHELL TYPE NEEDLE ROLLER BEARINGS








TA···Z TLA···Z　　TAM TLAM　　YT YTL

## Shaft dia. 12 — 15 mm

| Shaft dia. mm | Identification number | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Standard | Mass (Ref.) g | Closed end | Mass (Ref.) g | Standard | Mass (Ref.) g | Closed end | Mass (Ref.) g | Grease retained | Mass (Ref.) g |
| 12 | — | — | — | — | **TLA 1210 Z** | 4.3 | **TLAM 1210** | 4.7 | — | — |
| | — | — | — | — | — | — | — | — | **YTL 1210** | 5.1 |
| | — | — | — | — | **TLA 1212 Z** | 8.6 | **TLAM 1212** | 9.4 | — | — |
| | **TA 1212 Z** | 10.5 | **TAM 1212** | 11.5 | — | — | — | — | — | — |
| | **TA 1215 Z** | 13.1 | **TAM 1215** | 14 | — | — | — | — | — | — |
| | **TA 1220 Z** | 17.3 | **TAM 1220** | 18.3 | — | — | — | — | — | — |
| | **TA 1225 Z** | 21.5 | **TAM 1225** | 22.5 | — | — | — | — | — | — |
| | — | — | — | — | — | — | — | — | **YT 1212** | 12.8 |
| 13 | — | — | — | — | **TLA 1312 Z** | 9.2 | **TLAM 1312** | 10.1 | — | — |
| 14 | — | — | — | — | **TLA 1412 Z** | 9.8 | **TLAM 1412** | 10.8 | — | — |
| | — | — | — | — | **TLA 1416 Z** | 13.2 | **TLAM 1416** | 14.3 | — | — |
| | **TA 1416 Z** | 18.4 | **TAM 1416** | 19.6 | — | — | — | — | — | — |
| | **TA 1420 Z** | 23 | **TAM 1420** | 24 | — | — | — | — | — | — |
| 15 | — | — | — | — | **TLA 1512 Z** | 10.4 | **TLAM 1512** | 11.5 | — | — |
| | — | — | — | — | **TLA 1516 Z** | 14 | **TLAM 1516** | 15.2 | — | — |
| | — | — | — | — | **TLA 1522 Z** | 19.1 | **TLAM 1522** | 20.5 | — | — |
| | **TA 1510 Z** | 10.8 | **TAM 1510** | 12.3 | — | — | — | — | — | — |
| | **TA 1512 Z** | 12.9 | **TAM 1512** | 14.3 | — | — | — | — | — | — |
| | **TA 1515 Z** | 15.9 | **TAM 1515** | 17.3 | — | — | — | — | — | — |
| | **TA 1520 Z** | 21 | **TAM 1520** | 22.5 | — | — | — | — | — | — |
| | **TA 1525 Z** | 25 | **TAM 1525** | 26.5 | — | — | — | — | — | — |

| Boundary dimensions mm | | | | Standard mounting dimensions mm | | | | | | Basic dynamic load rating | Basic static load rating | Allowable rotational speed(1) | Assembled inner ring |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Shaft dia. h6 | | Housing bore dia. J7 | | Housing bore dia. N7 | | | | | |
| $F_w$ | $D$ | $C$ | $t_2$ Max. | Max. | Min. | Max. | Min. | Max. | Min. | $C$ N | $C_0$ N | rpm | |
| 12 | 16 | 10 | 1 | 12.000 | 11.989 | — | — | 15.995 | 15.977 | 4 350 | 5 810 | 35 000 | **IRT 810** |
| 12 | 16 | 10 | — | | | | | | | 7 470 | 11 800 | 13 000 | **IRT 810** |
| 12 | 18 | 12 | 1.3 | 12.000 | 11.989 | — | — | 17.995 | 17.977 | 6 420 | 7 490 | 35 000 | **IRT 812** |
| 12 | 19 | 12 | 1.3 | 12.000 | 11.989 | 19.012 | 18.991 | — | — | 6 000 | 6 310 | 35 000 | **IRT 812** |
| 12 | 19 | 15 | 1.3 | | | | | | | 7 440 | 8 320 | 35 000 | **IRT 815** |
| 12 | 19 | 20 | 1.3 | | | | | | | 10 700 | 13 300 | 35 000 | — |
| 12 | 19 | 25 | 1.3 | | | | | | | 13 800 | 18 300 | 35 000 | — |
| 12 | 19 | 12 | — | | | | | | | 11 800 | 15 200 | 13 000 | **IRT 812** |
| 13 | 19 | 12 | 1.3 | 13.000 | 12.989 | — | — | 18.993 | 18.972 | 6 760 | 8 170 | 30 000 | **IRT 1012** |
| 14 | 20 | 12 | 1.3 | 14.000 | 13.989 | — | — | 19.993 | 19.972 | 7 080 | 8 840 | 30 000 | **IRT 1012-2** |
| 14 | 20 | 16 | 1.3 | | | | | | | 8 950 | 12 000 | 30 000 | **IRT 1016-2** |
| 14 | 22 | 16 | 1.3 | 14.000 | 13.989 | 22.012 | 21.991 | — | — | 10 500 | 12 000 | 30 000 | **IRT 1016-2** |
| 14 | 22 | 20 | 1.3 | | | | | | | 13 900 | 17 200 | 30 000 | **IRT 1020-2** |
| 15 | 21 | 12 | 1.3 | 15.000 | 14.989 | — | — | 20.993 | 20.972 | 7 380 | 9 520 | 25 000 | **IRT 1212** |
| 15 | 21 | 16 | 1.3 | | | | | | | 9 330 | 12 900 | 25 000 | **IRT 1216** |
| 15 | 21 | 22 | 1.3 | | | | | | | 13 600 | 20 900 | 25 000 | **IRT 1222** |
| 15 | 22 | 10 | 1.3 | 15.000 | 14.989 | 22.012 | 21.991 | — | — | 5 290 | 5 680 | 25 000 | **IRT 1010-1** |
| 15 | 22 | 12 | 1.3 | | | | | | | 7 120 | 8 310 | 25 000 | **IRT 1012-1** |
| 15 | 22 | 15 | 1.3 | | | | | | | 8 830 | 11 000 | 25 000 | **IRT 1015-1** |
| 15 | 22 | 20 | 1.3 | | | | | | | 12 700 | 17 600 | 25 000 | **IRT 1020-1** |
| 15 | 22 | 25 | 1.3 | | | | | | | 16 300 | 24 200 | 25 000 | **IRT 1025-1** |

Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remark Shell Type Grease Retained Full Complement Needle Roller Bearings are provided with prepacked grease. Standard type and closed end type bearings are not provided with prepacked grease, so perform proper lubrication when using these types of bearings.

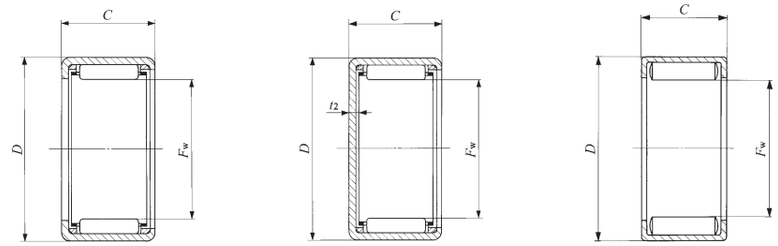


1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

# SHELL TYPE NEEDLE ROLLER BEARINGS








TA···Z TLA···Z　　TAM TLAM　　YT

## Shaft dia. 16 — 19mm

| Shaft dia. mm | Identification number | | | | | | | | | | Boundary dimensions mm | | | | Standard mounting dimensions mm | | | | | | Basic dynamic load rating $C$ N | Basic static load rating $C_0$ N | Allowable rotational speed(1) rpm | Assembled inner ring |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Standard | Mass (Ref.) g | Closed end | Mass (Ref.) g | Standard | Mass (Ref.) g | Closed end | Mass (Ref.) g | Grease retained | Mass (Ref.) g | $F_w$ | $D$ | $C$ | $t_2$ Max. | Shaft dia. h6 Max. | Shaft dia. h6 Min. | Housing bore dia. J7 Max. | Housing bore dia. J7 Min. | Housing bore dia. N7 Max. | Housing bore dia. N7 Min. | | | | |
| 16 | — | — | — | — | **TLA 1612 Z** | 10.9 | **TLAM 1612** | 12.2 | — | — | 16 | 22 | 12 | 1.3 | | | | | | | 7 670 | 10 200 | 25 000 | **IRT 1212-1** |
| | — | — | — | — | **TLA 1616 Z** | 14.8 | **TLAM 1616** | 16.1 | — | — | 16 | 22 | 16 | 1.3 | 16.000 | 15.989 | — | — | 21.993 | 21.972 | 9 700 | 13 800 | 25 000 | **IRT 1216-1** |
| | — | — | — | — | **TLA 1622 Z** | 20 | **TLAM 1622** | 21.5 | — | — | 16 | 22 | 22 | 1.3 | | | | | | | 14 200 | 22 400 | 25 000 | **IRT 1222-1** |
| | **TA 1616 Z** | 20 | **TAM 1616** | 22 | — | — | — | — | — | — | 16 | 24 | 16 | 1.3 | | | | | | | 11 100 | 13 300 | 25 000 | **IRT 1216-1** |
| | **TA 1620 Z** | 25 | **TAM 1620** | 27 | — | — | — | — | — | — | 16 | 24 | 20 | 1.3 | 16.000 | 15.989 | 24.012 | 23.991 | — | — | 14 700 | 19 100 | 25 000 | **IRT 1220-1** |
| 17 | — | — | — | — | **TLA 1712 Z** | 11.5 | **TLAM 1712** | 13 | — | — | 17 | 23 | 12 | 1.3 | 17.000 | 16.989 | — | — | 22.993 | 22.972 | 7 960 | 10 900 | 25 000 | — |
| | **TA 1715 Z** | 17.6 | **TAM 1715** | 19.5 | — | — | — | — | — | — | 17 | 24 | 15 | 1.3 | | | | | | | 9 660 | 12 700 | 25 000 | **IRT 1215-2** |
| | **TA 1720 Z** | 23.5 | **TAM 1720** | 25 | — | — | — | — | — | — | 17 | 24 | 20 | 1.3 | | | | | | | 13 900 | 20 400 | 25 000 | **IRT 1220-2** |
| | **TA 1725 Z** | 29 | **TAM 1725** | 31 | — | — | — | — | — | — | 17 | 24 | 25 | 1.3 | 17.000 | 16.989 | 24.012 | 23.991 | — | — | 17 900 | 28 100 | 25 000 | **IRT 1225-2** |
| | — | — | — | — | — | — | — | — | **YT 1715** | 20.5 | 17 | 24 | 15 | — | | | | | | | 16 600 | 26 000 | 9 000 | **IRT 1215-2** |
| | — | — | — | — | — | — | — | — | **YT 1725** | 35.5 | 17 | 24 | 25 | — | | | | | | | 27 200 | 49 000 | 9 000 | **IRT 1225-2** |
| 18 | — | — | — | — | **TLA 1812 Z** | 12 | **TLAM 1812** | 13.7 | — | — | 18 | 24 | 12 | 1.3 | 18.000 | 17.989 | — | — | 23.993 | 23.972 | 8 230 | 11 500 | 20 000 | **IRT 1512** |
| | — | — | — | — | **TLA 1816 Z** | 16.2 | **TLAM 1816** | 17.9 | — | — | 18 | 24 | 16 | 1.3 | | | | | | | 10 400 | 15 600 | 20 000 | **IRT 1516** |
| | **TA 1813 Z** | 16.4 | **TAM 1813** | 18.5 | — | — | — | — | — | — | 18 | 25 | 13 | 1.3 | | | | | | | 9 100 | 12 000 | 20 000 | **IRT 1513** |
| | **TA 1815 Z** | 18.5 | **TAM 1815** | 20.5 | — | — | — | — | — | — | 18 | 25 | 15 | 1.3 | | | | | | | 10 100 | 13 600 | 20 000 | **IRT 1515** |
| | **TA 1817 Z** | 21 | **TAM 1817** | 23 | — | — | — | — | — | — | 18 | 25 | 17 | 1.3 | | | | | | | 11 900 | 16 900 | 20 000 | **IRT 1517** |
| | **TA 1819 Z** | 23.5 | **TAM 1819** | 25.5 | — | — | — | — | — | — | 18 | 25 | 19 | 1.3 | 18.000 | 17.989 | 25.012 | 24.991 | — | — | 13 700 | 20 200 | 20 000 | **IRT 1519** |
| | **TA 1820 Z** | 24.5 | **TAM 1820** | 26.5 | — | — | — | — | — | — | 18 | 25 | 20 | 1.3 | | | | | | | 14 500 | 21 800 | 20 000 | **IRT 1520** |
| | **TA 1825 Z** | 30.5 | **TAM 1825** | 32.5 | — | — | — | — | — | — | 18 | 25 | 25 | 1.3 | | | | | | | 18 600 | 30 000 | 20 000 | **IRT 1525** |
| 19 | **TA 1916 Z** | 23 | **TAM 1916** | 25.5 | — | — | — | — | — | — | 19 | 27 | 16 | 1.3 | 19.000 | 18.987 | 27.012 | 26.991 | — | — | 12 200 | 15 700 | 20 000 | **IRT 1516-1** |
| | **TA 1920 Z** | 29 | **TAM 1920** | 31 | — | — | — | — | — | — | 19 | 27 | 20 | 1.3 | | | | | | | 16 100 | 22 600 | 20 000 | **IRT 1520-1** |

Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remark Shell Type Grease Retained Full Complement Needle Roller Bearings are provided with prepacked grease. Standard type and closed end type bearings are not provided with prepacked grease, so perform proper lubrication when using these types of bearings.

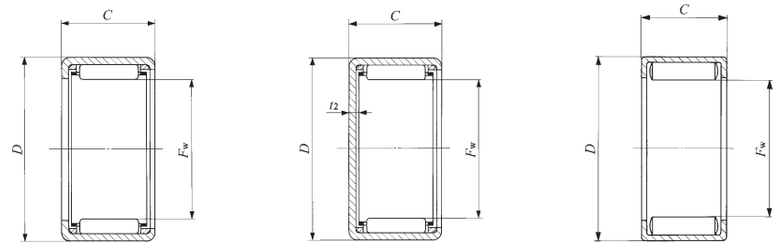


1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

# SHELL TYPE NEEDLE ROLLER BEARINGS








TA···Z TLA···Z　　TAM TLAM　　YT

## Shaft dia. 20 — 21mm

| Shaft dia. mm | Identification number | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Standard | Mass (Ref.) g | Closed end | Mass (Ref.) g | Standard | Mass (Ref.) g | Closed end | Mass (Ref.) g | Grease retained | Mass (Ref.) g |
| 20 | — | — | — | — | **TLA 2012 Z** | 13.2 | **TLAM 2012** | 15.2 | — | — |
| | — | — | — | — | **TLA 2016 Z** | 17.8 | **TLAM 2016** | 19.9 | — | — |
| | — | — | — | — | **TLA 2020 Z** | 22 | **TLAM 2020** | 24 | — | — |
| | — | — | — | — | **TLA 2030 Z** | 33 | **TLAM 2030** | 35 | — | — |
| | **TA 2015 Z** | 20 | **TAM 2015** | 22.5 | — | — | — | — | — | — |
| | **TA 2020 Z** | 26.5 | **TAM 2020** | 29 | — | — | — | — | — | — |
| | **TA 2025 Z** | 33 | **TAM 2025** | 35.5 | — | — | — | — | — | — |
| | **TA 2030 Z** | 39.5 | **TAM 2030** | 42 | — | — | — | — | — | — |
| | — | — | — | — | — | — | — | — | **YT 2015** | 23.5 |
| | — | — | — | — | — | — | — | — | **YT 2025** | 41 |
| | **TA 202820 Z** | 30 | **TAM 202820** | 32.5 | — | — | — | — | — | — |
| | — | — | — | — | — | — | — | — | **YT 202820** | 37.5 |
| 21 | **TA 2116 Z** | 25 | **TAM 2116** | 28 | — | — | — | — | — | — |
| | **TA 2120 Z** | 31.5 | **TAM 2120** | 34.5 | — | — | — | — | — | — |
| | — | — | — | — | — | — | — | — | **YT 2116** | 31 |
| | — | — | — | — | — | — | — | — | **YT 2120** | 39 |

| Boundary dimensions mm | | | | Standard mounting dimensions mm | | | | | | Basic dynamic load rating | Basic static load rating | Allowable rotational speed(1) | Assembled inner ring |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Shaft dia. h6 | | Housing bore dia. J7 | | Housing bore dia. N7 | | | | | |
| $F_w$ | $D$ | $C$ | $t_2$ Max. | Max. | Min. | Max. | Min. | Max. | Min. | $C$ N | $C_0$ N | rpm | |
| 20 | 26 | 12 | 1.3 | 20.000 | 19.987 | — | — | 25.993 | 25.972 | 8 740 | 12 900 | 20 000 | — |
| 20 | 26 | 16 | 1.3 | | | | | | | 11 100 | 17 500 | 20 000 | **IRT 1716** |
| 20 | 26 | 20 | 1.3 | | | | | | | 14 500 | 24 700 | 20 000 | **IRT 1720** |
| 20 | 26 | 30 | 1.3 | | | | | | | 22 300 | 42 900 | 20 000 | **IRT 1730** |
| 20 | 27 | 15 | 1.3 | 20.000 | 19.987 | 27.012 | 26.991 | — | — | 10 400 | 14 600 | 20 000 | **IRT 1515-2** |
| 20 | 27 | 20 | 1.3 | | | | | | | 15 000 | 23 400 | 20 000 | **IRT 1520-2** |
| 20 | 27 | 25 | 1.3 | | | | | | | 19 200 | 32 200 | 20 000 | **IRT 1525-2** |
| 20 | 27 | 30 | 1.3 | | | | | | | 23 100 | 41 000 | 20 000 | **IRT 1530-2** |
| 20 | 27 | 15 | — | | | | | | | 18 400 | 30 900 | 7 500 | **IRT 1515-2** |
| 20 | 27 | 25 | — | | | | | | | 30 000 | 58 300 | 7 500 | **IRT 1525-2** |
| 20 | 28 | 20 | 1.3 | 20.000 | 19.987 | 28.012 | 27.991 | — | — | 16 900 | 24 300 | 20 000 | **IRT 1520-2** |
| 20 | 28 | 20 | — | | | | | | | 26 800 | 44 600 | 7 500 | **IRT 1520-2** |
| 21 | 29 | 16 | 1.3 | 21.000 | 20.987 | 29.012 | 28.991 | — | — | 13 300 | 18 100 | 19 000 | **IRT 1716-1** |
| 21 | 29 | 20 | 1.3 | | | | | | | 17 600 | 25 900 | 19 000 | **IRT 1720-1** |
| 21 | 29 | 16 | — | | | | | | | 22 100 | 35 200 | 7 000 | **IRT 1716-1** |
| 21 | 29 | 20 | — | | | | | | | 27 500 | 46 800 | 7 000 | **IRT 1720-1** |

Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remark Shell Type Grease Retained Full Complement Needle Roller Bearings are provided with prepacked grease. Standard type and closed end type bearings are not provided with prepacked grease, so perform proper lubrication when using these types of bearings.



1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

# SHELL TYPE NEEDLE ROLLER BEARINGS

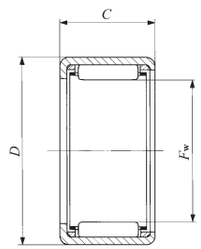




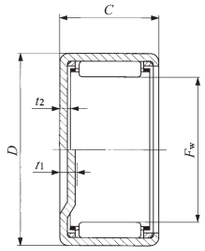
## Shaft dia. 22 — 24mm

| Shaft dia. mm | Identification number | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Standard | Mass (Ref.) g | Closed end | Mass (Ref.) g | Standard | Mass (Ref.) g | Closed end | Mass (Ref.) g | Grease retained | Mass (Ref.) g |
| 22 | — | — | — | — | **TLA 2212 Z** | 15.6 | **TLAM 2212** | 18.1 | — | — |
| | — | — | — | — | **TLA 2216 Z** | 21.5 | **TLAM 2216** | 24 | — | — |
| | — | — | — | — | **TLA 2220 Z** | 26.5 | **TLAM 2220** | 29 | — | — |
| | **TA 2210 Z** | 15 | **TAM 2210** | 18.1 | — | — | — | — | — | — |
| | **TA 2215 Z** | 21.5 | **TAM 2215** | 24.5 | — | — | — | — | — | — |
| | **TA 2220 Z** | 29 | **TAM 2220** | 32 | — | — | — | — | — | — |
| | **TA 2225 Z** | 35.5 | **TAM 2225** | 38.5 | — | — | — | — | — | — |
| | **TA 2230 Z** | 42.5 | **TAM 2230** | 45.5 | — | — | — | — | — | — |
| | **TA 223016 Z** | 26 | **TAM 223016** | 29 | — | — | — | — | — | — |
| | **TA 223020 Z** | 32.5 | **TAM 223020** | 35.5 | — | — | — | — | — | — |
| | — | — | — | — | — | — | — | — | **YT 223016** | 32 |
| | — | — | — | — | — | — | — | — | **YT 223020** | 40.5 |
| 24 | **TA 2420 Z** | 31 | **TAM 2420** | 35 | — | — | — | — | — | — |
| | **TA 2428 Z** | 43.5 | **TAM 2428** | 47 | — | — | — | — | — | — |
| | — | — | — | — | — | — | — | — | **YT 2428** | 54 |
| | **TA 243216 Z** | 28 | **TAM 243216** | 32 | — | — | — | — | — | — |
| | **TA 243220 Z** | 35.5 | **TAM 243220** | 39 | — | — | — | — | — | — |
| | — | — | — | — | — | — | — | — | **YT 243216** | 34.5 |
| | — | — | — | — | — | — | — | — | **YT 243220** | 43.5 |

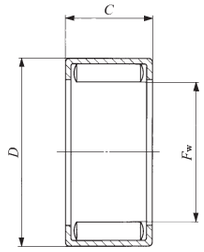
Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remark Shell Type Grease Retained Full Complement Needle Roller Bearings are provided with prepacked grease. Standard type and closed end type bearings are not provided with prepacked grease, so perform proper lubrication when using these types of bearings.




TA···Z TLA···Z




TAM TLAM
$t_1$ ($F_w \geqq 24$)
$t_2$ ($F_w \leqq 22$)




YT

| Boundary dimensions mm | | | | Standard mounting dimensions mm | | | | | | Basic dynamic load rating | Basic static load rating | Allowable rotational speed(1) | Assembled inner ring |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Shaft dia. h6 | | Housing bore dia. J7 | | N7 | | | | | |
| $F_w$ | $D$ | $C$ | $t_1$, $t_2$ Max. | Max. | Min. | Max. | Min. | Max. | Min. | $C$ N | $C_0$ N | rpm | |
| 22 | 28 | 12 | 1.3 | | | | | | | 9 230 | 14 300 | 18 000 | — |
| 22 | 28 | 16 | 1.3 | 22.000 | 21.987 | — | — | 27.993 | 27.972 | 11 700 | 19 300 | 18 000 | **IRT 1716-2** |
| 22 | 28 | 20 | 1.3 | | | | | | | 15 300 | 27 300 | 18 000 | **IRT 1720-2** |
| 22 | 29 | 10 | 1.3 | | | | | | | 6 650 | 8 500 | 18 000 | **IRT 1710-2** |
| 22 | 29 | 15 | 1.3 | | | | | | | 11 100 | 16 400 | 18 000 | **IRT 1715-2** |
| 22 | 29 | 20 | 1.3 | 22.000 | 21.987 | 29.012 | 28.991 | — | — | 16 000 | 26 300 | 18 000 | **IRT 1720-2** |
| 22 | 29 | 25 | 1.3 | | | | | | | 19 700 | 34 300 | 18 000 | **IRT 1725-2** |
| 22 | 29 | 30 | 1.3 | | | | | | | 23 800 | 43 700 | 18 000 | **IRT 1730-2** |
| 22 | 30 | 16 | 1.3 | | | | | | | 13 200 | 18 200 | 18 000 | **IRT 1716-2** |
| 22 | 30 | 20 | 1.3 | 22.000 | 21.987 | 30.012 | 29.991 | — | — | 17 500 | 26 100 | 18 000 | **IRT 1720-2** |
| 22 | 30 | 16 | — | | | | | | | 22 600 | 36 800 | 7 000 | **IRT 1716-2** |
| 22 | 30 | 20 | — | | | | | | | 28 200 | 48 900 | 7 000 | **IRT 1720-2** |
| 24 | 31 | 20 | 3.4 | | | | | | | 17 000 | 29 200 | 16 000 | **IRT 2020** |
| 24 | 31 | 28 | 3.4 | 24.000 | 23.987 | 31.014 | 30.989 | — | — | 24 500 | 46 700 | 16 000 | **IRT 2028** |
| 24 | 31 | 28 | — | | | | | | | 36 800 | 79 900 | 6 500 | **IRT 2028** |
| 24 | 32 | 16 | 3.4 | | | | | | | 14 200 | 20 500 | 16 000 | **IRT 2016** |
| 24 | 32 | 20 | 3.4 | 24.000 | 23.987 | 32.014 | 31.989 | — | — | 18 800 | 29 400 | 16 000 | **IRT 2020** |
| 24 | 32 | 16 | — | | | | | | | 23 700 | 40 100 | 6 500 | **IRT 2016** |
| 24 | 32 | 20 | — | | | | | | | 29 500 | 53 200 | 6 500 | **IRT 2020** |

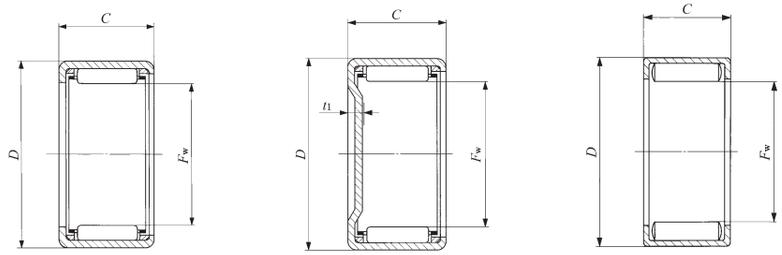


1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

# SHELL TYPE NEEDLE ROLLER BEARINGS








TA···Z TLA···Z　　TAM TLAM　　YT YTL

## Shaft dia. 25 — 28 mm

| Shaft dia. mm | Identification number | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Standard | Mass (Ref.) g | Closed end | Mass (Ref.) g | Standard | Mass (Ref.) g | Closed end | Mass (Ref.) g | Grease retained | Mass (Ref.) g |
| 25 | — | — | — | — | **TLA 2512 Z** | 19.7 | **TLAM 2512** | 23.5 | — | — |
| | — | — | — | — | **TLA 2516 Z** | 26 | **TLAM 2516** | 29.5 | — | — |
| | — | — | — | — | **TLA 2520 Z** | 32 | **TLAM 2520** | 36 | — | — |
| | — | — | — | — | **TLA 2526 Z** | 41.5 | **TLAM 2526** | 45.5 | — | — |
| | — | — | — | — | **TLAW2538Z** | 58.5 | **TLAMW2538** | 62 | — | — |
| | — | — | — | — | — | — | — | — | **YTL 2526** | 51.5 |
| | **TA 2510 Z** | 19.1 | **TAM 2510** | 23 | — | — | — | — | — | — |
| | **TA 2515 Z** | 28.5 | **TAM 2515** | 32.5 | — | — | — | — | — | — |
| | **TA 2520 Z** | 36.5 | **TAM 2520** | 40.5 | — | — | — | — | — | — |
| | **TA 2525 Z** | 45.5 | **TAM 2525** | 49 | — | — | — | — | — | — |
| | **TA 2530 Z** | 54.5 | **TAM 2530** | 58.5 | — | — | — | — | — | — |
| | — | — | — | — | — | — | — | — | **YT 2510** | 22.5 |
| | — | — | — | — | — | — | — | — | **YT 2515** | 33 |
| | — | — | — | — | — | — | — | — | **YT 2520** | 45 |
| | — | — | — | — | — | — | — | — | **YT 2525** | 57 |
| 26 | **TA 2616 Z** | 30.5 | **TAM 2616** | 34.5 | — | — | — | — | — | — |
| | **TA 2620 Z** | 38 | **TAM 2620** | 42.5 | — | — | — | — | — | — |
| | — | — | — | — | — | — | — | — | **YT 2616** | 37 |
| | — | — | — | — | — | — | — | — | **YT 2620** | 46.5 |
| 28 | — | — | — | — | **TLA 2816 Z** | 28.5 | **TLAM 2816** | 33.5 | — | — |
| | — | — | — | — | **TLA 2820 Z** | 35.5 | **TLAM 2820** | 40.5 | — | — |
| | **TA 2820 Z** | 45 | **TAM 2820** | 50 | — | — | — | — | — | — |
| | **TA 2830 Z** | 67.5 | **TAM 2830** | 72.5 | — | — | — | — | — | — |
| | — | — | — | — | — | — | — | — | **YT 2820** | 56.5 |

| Boundary dimensions mm | | | | Standard mounting dimensions mm | | | | | | Basic dynamic load rating | Basic static load rating | Allowable rotational speed(1) | Assembled inner ring |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Shaft dia. h6 | | Housing bore dia. J7 | | Housing bore dia. N7 | | | | | |
| $F_w$ | $D$ | $C$ | $t_1$ Max. | Max. | Min. | Max. | Min. | Max. | Min. | $C$ N | $C_0$ N | rpm | |
| 25 | 32 | 12 | 2.8 | 25.000 | 24.987 | — | — | 31.992 | 31.967 | 9 440 | 13 900 | 15 000 | — |
| 25 | 32 | 16 | 2.8 | | | | | | | 12 800 | 20 500 | 15 000 | — |
| 25 | 32 | 20 | 2.8 | | | | | | | 16 900 | 29 300 | 15 000 | **IRT 2020-1** |
| 25 | 32 | 26 | 2.8 | | | | | | | 22 600 | 42 500 | 15 000 | **IRT 2026-1** |
| 25 | 32 | 38 | 2.8 | | | | | | | 28 900 | 58 500 | 15 000 | **IRT 2038-1** |
| 25 | 32 | 26 | — | | | | | | | 35 000 | 75 800 | 6 000 | **IRT 2026-1** |
| 25 | 33 | 10 | 3.4 | 25.000 | 24.987 | 33.014 | 32.989 | — | — | 7 990 | 9 900 | 15 000 | **IRT 2010-1** |
| 25 | 33 | 15 | 3.4 | | | | | | | 13 400 | 19 300 | 15 000 | **IRT 2015-1** |
| 25 | 33 | 20 | 3.4 | | | | | | | 19 500 | 31 100 | 15 000 | **IRT 2020-1** |
| 25 | 33 | 25 | 3.4 | | | | | | | 24 100 | 40 800 | 15 000 | **IRT 2025-1** |
| 25 | 33 | 30 | 3.4 | | | | | | | 29 100 | 52 000 | 15 000 | **IRT 2030-1** |
| 25 | 33 | 10 | — | 25.000 | 24.987 | 33.014 | 32.989 | — | — | 15 500 | 23 600 | 6 000 | **IRT 2010-1** |
| 25 | 33 | 15 | — | | | | | | | 22 700 | 38 300 | 6 000 | **IRT 2015-1** |
| 25 | 33 | 20 | — | | | | | | | 30 200 | 55 400 | 6 000 | **IRT 2020-1** |
| 25 | 33 | 25 | — | | | | | | | 37 200 | 72 500 | 6 000 | **IRT 2025-1** |
| 26 | 34 | 16 | 3.4 | 26.000 | 25.987 | 34.014 | 33.989 | — | — | 15 200 | 22 900 | 15 000 | **IRT 2216** |
| 26 | 34 | 20 | 3.4 | | | | | | | 20 100 | 32 800 | 15 000 | **IRT 2220** |
| 26 | 34 | 16 | — | | | | | | | 24 700 | 43 300 | 6 000 | **IRT 2216** |
| 26 | 34 | 20 | — | | | | | | | 30 800 | 57 500 | 6 000 | **IRT 2220** |
| 28 | 35 | 16 | 2.8 | 28.000 | 27.987 | — | — | 34.992 | 34.967 | 13 800 | 23 500 | 13 000 | — |
| 28 | 35 | 20 | 2.8 | | | | | | | 18 300 | 33 600 | 13 000 | **IRT 2220-1** |
| 28 | 37 | 20 | 3.4 | 28.000 | 27.987 | 37.014 | 36.989 | — | — | 21 200 | 32 300 | 13 000 | **IRT 2220-1** |
| 28 | 37 | 30 | 3.4 | | | | | | | 33 000 | 56 900 | 13 000 | **IRT 2230-1** |
| 28 | 37 | 20 | — | | | | | | | 34 700 | 61 700 | 5 500 | **IRT 2220-1** |

Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.

Remarks1. "W" in the identification number indicates that rolling elements are arranged in double rows.

2. Shell Type Grease Retained Full Complement Needle Roller Bearings are provided with prepacked grease. Standard type and closed end type bearings are not provided with prepacked grease, so perform proper lubrication when using these types of bearings.

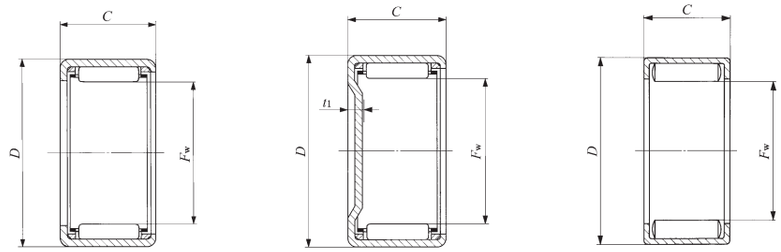


1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

# SHELL TYPE NEEDLE ROLLER BEARINGS








TA···Z TLA···Z　　TAM TLAM　　YT

## Shaft dia. 29 — 35mm

| Shaft dia. mm | Identification number | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Standard | Mass (Ref.) g | Closed end | Mass (Ref.) g | Standard | Mass (Ref.) g | Closed end | Mass (Ref.) g | Grease retained | Mass (Ref.) g |
| 29 | **TA 2920 Z** | 47 | **TAM 2920** | 52 | — | — | — | — | — | — |
| | **TA 2930 Z** | 70 | **TAM 2930** | 75.5 | — | — | — | — | — | — |
| | — | — | — | — | — | — | — | — | **YT 2920** | 58.5 |
| 30 | — | — | — | — | **TLA 3012 Z** | 23.5 | **TLAM 3012** | 29 | — | — |
| | — | — | — | — | **TLA 3016 Z** | 30.5 | **TLAM 3016** | 36 | — | — |
| | — | — | — | — | **TLA 3018 Z** | 34.5 | **TLAM 3018** | 40 | — | — |
| | — | — | — | — | **TLA 3020 Z** | 38 | **TLAM 3020** | 43.5 | — | — |
| | — | — | — | — | **TLA 3026 Z** | 49 | **TLAM 3026** | 54.5 | — | — |
| | — | — | — | — | **TLAW3038 Z** | 69 | **TLAMW3038** | 74.5 | — | — |
| | **TA 3013 Z** | 36.5 | **TAM 3013** | 42.5 | — | — | — | — | — | — |
| | **TA 3015 Z** | 42 | **TAM 3015** | 47.5 | — | — | — | — | — | — |
| | **TA 3020 Z** | 54.5 | **TAM 3020** | 60 | — | — | — | — | — | — |
| | **TA 3025 Z** | 68 | **TAM 3025** | 73.5 | — | — | — | — | — | — |
| | **TA 3030 Z** | 80 | **TAM 3030** | 85.5 | — | — | — | — | — | — |
| 32 | **TA 3220 Z** | 57.5 | **TAM 3220** | 63.5 | — | — | — | — | — | — |
| | **TA 3230 Z** | 86 | **TAM 3230** | 97.5 | — | — | — | — | — | — |
| | — | — | — | — | — | — | — | — | **YT 3220** | 71.5 |
| 35 | — | — | — | — | **TLA 3512 Z** | 27 | **TLAM 3512** | 34.5 | — | — |
| | — | — | — | — | **TLA 3516 Z** | 35 | **TLAM 3516** | 42.5 | — | — |
| | — | — | — | — | **TLA 3520 Z** | 43.5 | **TLAM 3520** | 51 | — | — |
| | **TA 3512 Z** | 38.5 | **TAM 3512** | 46 | — | — | — | — | — | — |
| | **TA 3515 Z** | 48 | **TAM 3515** | 56 | — | — | — | — | — | — |
| | **TA 3520 Z** | 62.5 | **TAM 3520** | 70 | — | — | — | — | — | — |
| | **TA 3525 Z** | 78 | **TAM 3525** | 85.5 | — | — | — | — | — | — |
| | **TA 3530 Z** | 97 | **TAM 3530** | 105 | — | — | — | — | — | — |

| Boundary dimensions mm | | | | Standard mounting dimensions mm | | | | | | Basic dynamic load rating | Basic static load rating | Allowable rotational speed(1) | Assembled inner ring |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Shaft dia. h6 | | Housing bore dia. J7 | | N7 | | | | | |
| $F_w$ | $D$ | $C$ | $t_1$ Max. | Max. | Min. | Max. | Min. | Max. | Min. | $C$ N | $C_0$ N | rpm | |
| 29 | 38 | 20 | 3.4 | | | | | | | 22 000 | 34 200 | 13 000 | **IRT 2520** |
| 29 | 38 | 30 | 3.4 | 29.000 | 28.987 | 38.014 | 37.989 | — | — | 34 200 | 60 300 | 13 000 | **IRT 2530** |
| 29 | 38 | 20 | — | | | | | | | 35 500 | 64 100 | 5 000 | **IRT 2520** |
| 30 | 37 | 12 | 2.8 | | | | | | | 10 400 | 16 600 | 12 000 | — |
| 30 | 37 | 16 | 2.8 | | | | | | | 14 100 | 24 500 | 12 000 | — |
| 30 | 37 | 18 | 2.8 | | | | | | | 16 400 | 29 800 | 12 000 | — |
| 30 | 37 | 20 | 2.8 | 30.000 | 29.987 | — | — | 36.992 | 36.967 | 18 600 | 35 100 | 12 000 | **IRT 2520-1** |
| 30 | 37 | 26 | 2.8 | | | | | | | 24 800 | 50 900 | 12 000 | **IRT 2526-1** |
| 30 | 37 | 38 | 2.8 | | | | | | | 31 900 | 70 200 | 12 000 | **IRT 2538-1** |
| 30 | 40 | 13 | 3.4 | | | | | | | 13 500 | 16 800 | 12 000 | — |
| 30 | 40 | 15 | 3.4 | | | | | | | 16 800 | 22 400 | 12 000 | **IRT 2515-1** |
| 30 | 40 | 20 | 3.4 | 30.000 | 29.987 | 40.014 | 39.989 | — | — | 24 500 | 36 300 | 12 000 | **IRT 2520-1** |
| 30 | 40 | 25 | 3.4 | | | | | | | 31 600 | 50 300 | 12 000 | **IRT 2525-1** |
| 30 | 40 | 30 | 3.4 | | | | | | | 36 700 | 60 700 | 12 000 | **IRT 2530-1** |
| 32 | 42 | 20 | 3.4 | | | | | | | 25 400 | 38 600 | 11 000 | **IRT 2820** |
| 32 | 42 | 30 | 3.4 | 32.000 | 31.984 | 42.014 | 41.989 | — | — | 39 500 | 68 400 | 11 000 | **IRT 2830** |
| 32 | 42 | 20 | — | | | | | | | 39 900 | 70 100 | 4 500 | **IRT 2820** |
| 35 | 42 | 12 | 2.8 | | | | | | | 11 600 | 20 000 | 10 000 | **IRT 3012** |
| 35 | 42 | 16 | 2.8 | 35.000 | 34.984 | — | — | 41.992 | 41.967 | 15 700 | 29 600 | 10 000 | — |
| 35 | 42 | 20 | 2.8 | | | | | | | 20 700 | 42 300 | 10 000 | **IRT 3020** |
| 35 | 45 | 12 | 3.4 | | | | | | | 14 800 | 19 900 | 10 000 | **IRT 3012** |
| 35 | 45 | 15 | 3.4 | | | | | | | 18 500 | 26 500 | 10 000 | **IRT 3015** |
| 35 | 45 | 20 | 3.4 | 35.000 | 34.984 | 45.014 | 44.989 | — | — | 27 000 | 43 100 | 10 000 | **IRT 3020** |
| 35 | 45 | 25 | 3.4 | | | | | | | 34 800 | 59 700 | 10 000 | **IRT 3025** |
| 35 | 45 | 30 | 3.4 | | | | | | | 40 600 | 72 600 | 10 000 | **IRT 3030** |

Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. "W" in the identification number indicates that rolling elements are arranged in double rows.
2. Shell Type Grease Retained Full Complement Needle Roller Bearings are provided with prepacked grease. Standard type and closed end type bearings are not provided with prepacked grease, so perform proper lubrication when using these types of bearings.

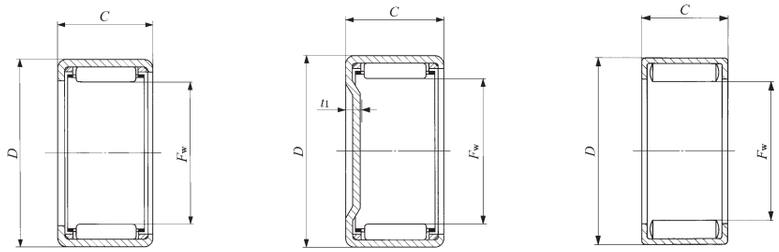


1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

# SHELL TYPE NEEDLE ROLLER BEARINGS








TA···Z TLA···Z TAM TLAM YT

## Shaft dia. 37 — 45mm

| Shaft dia. mm | Identification number | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Standard | Mass (Ref.) g | Closed end | Mass (Ref.) g | Standard | Mass (Ref.) g | Closed end | Mass (Ref.) g | Grease retained | Mass (Ref.) g |
| 37 | **TA 3720 Z** | 64.5 | **TAM 3720** | 73 | — | — | — | — | — | — |
| | **TA 3730 Z** | 101 | **TAM 3730** | 110 | — | — | — | — | — | — |
| | — | — | — | — | — | — | — | — | **YT 3720** | 81 |
| 38 | **TA 3815 Z** | 51 | **TAM 3815** | 60 | — | — | — | — | — | — |
| | **TA 3820 Z** | 65.5 | **TAM 3820** | 74.5 | — | — | — | — | — | — |
| | **TA 3825 Z** | 82.5 | **TAM 3825** | 96 | — | — | — | — | — | — |
| | **TA 3830 Z** | 104 | **TAM 3830** | 114 | — | — | — | — | — | — |
| | **TAW 3845 Z** | 149 | **TAMW 3845** | 159 | — | — | — | — | — | — |
| 40 | — | — | — | — | **TLA 4012 Z** | 30 | **TLAM 4012** | 40 | — | — |
| | — | — | — | — | **TLA 4016 Z** | 39 | **TLAM 4016** | 49 | — | — |
| | — | — | — | — | **TLA 4020 Z** | 49 | **TLAM 4020** | 58.5 | — | — |
| | **TA 4015 Z** | 54 | **TAM 4015** | 63.5 | — | — | — | — | — | — |
| | **TA 4020 Z** | 69.5 | **TAM 4020** | 79 | — | — | — | — | — | — |
| | **TA 4025 Z** | 86.5 | **TAM 4025** | 102 | — | — | — | — | — | — |
| | **TA 4030 Z** | 110 | **TAM 4030** | 120 | — | — | — | — | — | — |
| | **TA 4040 Z** | 144 | **TAM 4040** | 154 | — | — | — | — | — | — |
| | — | — | — | — | — | — | — | — | **YT 4015** | 63.5 |
| | — | — | — | — | — | — | — | — | **YT 4025** | 109 |
| 45 | — | — | — | — | **TLA 4516 Z** | 43.5 | **TLAM 4516** | 56 | — | — |
| | — | — | — | — | **TLA 4520 Z** | 54.5 | **TLAM 4520** | 67 | — | — |
| | **TA 4520 Z** | 77 | **TAM 4520** | 90 | — | — | — | — | — | — |
| | **TA 4525 Z** | 102 | **TAM 4525** | 115 | — | — | — | — | — | — |
| | **TA 4530 Z** | 122 | **TAM 4530** | 135 | — | — | — | — | — | — |
| | **TA 4540 Z** | 161 | **TAM 4540** | 174 | — | — | — | — | — | — |
| | — | — | — | — | — | — | — | — | **YT 4520** | 96 |
| | — | — | — | — | — | — | — | — | **YT 4525** | 122 |

| Boundary dimensions mm | | | | Standard mounting dimensions mm | | | | | | Basic dynamic load rating | Basic static load rating | Allowable rotational speed(1) | Assembled inner ring |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Shaft dia. h6 | | Housing bore dia. J7 | | N7 | | | | | |
| $F_w$ | $D$ | $C$ | $t_1$ Max. | Max. | Min. | Max. | Min. | Max. | Min. | $C$ N | $C_0$ N | rpm | |
| 37 | 47 | 20 | 3.4 | | | | | | | 27 800 | 45 400 | 9 500 | **IRT 3220** |
| 37 | 47 | 30 | 3.4 | 37.000 | 36.984 | 47.014 | 46.989 | — | — | 41 800 | 76 700 | 9 500 | **IRT 3230** |
| 37 | 47 | 20 | — | | | | | | | 43 300 | 81 300 | 4 000 | **IRT 3220** |
| 38 | 48 | 15 | 3.4 | | | | | | | 19 000 | 28 000 | 9 000 | **IRT 3215-1** |
| 38 | 48 | 20 | 3.4 | | | | | | | 27 700 | 45 600 | 9 000 | **IRT 3220-1** |
| 38 | 48 | 25 | 3.4 | 38.000 | 37.984 | 48.014 | 47.989 | — | — | 35 600 | 63 100 | 9 000 | **IRT 3225-1** |
| 38 | 48 | 30 | 3.4 | | | | | | | 43 100 | 80 600 | 9 000 | **IRT 3230-1** |
| 38 | 48 | 45 | 3.4 | | | | | | | 55 700 | 112 000 | 9 000 | **IRT 3245-1** |
| 40 | 47 | 12 | 2.8 | | | | | | | 12 400 | 22 800 | 8 500 | — |
| 40 | 47 | 16 | 2.8 | 40.000 | 39.984 | — | — | 46.992 | 46.967 | 16 700 | 33 700 | 8 500 | — |
| 40 | 47 | 20 | 2.8 | | | | | | | 22 100 | 48 200 | 8 500 | **IRT 3520** |
| 40 | 50 | 15 | 3.4 | | | | | | | 19 500 | 29 400 | 8 500 | **IRT 3515** |
| 40 | 50 | 20 | 3.4 | | | | | | | 28 400 | 47 800 | 8 500 | **IRT 3520** |
| 40 | 50 | 25 | 3.4 | | | | | | | 36 600 | 66 200 | 8 500 | **IRT 3525** |
| 40 | 50 | 30 | 3.4 | 40.000 | 39.984 | 50.014 | 49.989 | — | — | 44 300 | 84 600 | 8 500 | **IRT 3530** |
| 40 | 50 | 40 | 3.4 | | | | | | | 56 700 | 116 000 | 8 500 | **IRT 3540** |
| 40 | 50 | 15 | — | | | | | | | 33 400 | 59 800 | 4 000 | **IRT 3515** |
| 40 | 50 | 25 | — | | | | | | | 55 300 | 114 000 | 4 000 | **IRT 3525** |
| 45 | 52 | 16 | 2.8 | 45.000 | 44.984 | — | — | 51.991 | 51.961 | 17 800 | 37 800 | 7 500 | — |
| 45 | 52 | 20 | 2.8 | | | | | | | 23 400 | 54 000 | 7 500 | **IRT 4020** |
| 45 | 55 | 20 | 3.4 | | | | | | | 30 600 | 54 600 | 7 500 | **IRT 4020** |
| 45 | 55 | 25 | 3.4 | | | | | | | 39 400 | 75 600 | 7 500 | **IRT 4025** |
| 45 | 55 | 30 | 3.4 | 45.000 | 44.984 | 55.018 | 54.988 | — | — | 47 700 | 96 600 | 7 500 | **IRT 4030** |
| 45 | 55 | 40 | 3.4 | | | | | | | 61 300 | 133 000 | 7 500 | **IRT 4040** |
| 45 | 55 | 20 | — | | | | | | | 47 800 | 98 200 | 3 500 | **IRT 4020** |
| 45 | 55 | 25 | — | | | | | | | 59 100 | 129 000 | 3 500 | **IRT 4025** |

Note(1) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.

Remarks1. "W" in the identification number indicates that rolling elements are arranged in double rows.

2. Shell Type Grease Retained Full Complement Needle Roller Bearings are provided with prepacked grease. Standard type and closed end type bearings are not provided with prepacked grease, so perform proper lubrication when using these types of bearings.



1N=0.102kgf=0.2248lbs.
1mm=0.03937inch